

Online Course Management System Architecture (MVC)

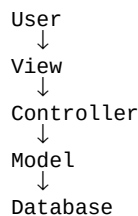
Introduction

This document presents the architectural design of an Online Course Management System supporting students, instructors, and admins with focus on scalability, maintainability, security, usability, and data persistence.

Architectural Style: Model-View-Controller (MVC)

MVC separates the system into Model, View, and Controller, improving maintainability and scalability. It allows independent UI development and better organization of business logic and data handling.

High-Level Design



Key Components

View: UI layer

Controller: Handles logic

Model: Manages data

Database: Stores persistent data

Design Decisions

- Relational Database (MySQL)
- JWT Authentication
- REST API communication

Trade-offs

- Increased complexity
- Slower initial development
- Controller overload (mitigated by modular controllers)

Design Patterns

- Factory Pattern (user creation)
- Observer Pattern (notifications)
- Singleton Pattern (database connection)

Reflection

This design highlights the importance of structuring systems before coding. MVC was chosen over microservices due to its simplicity and suitability for initial development while still supporting future scalability.